

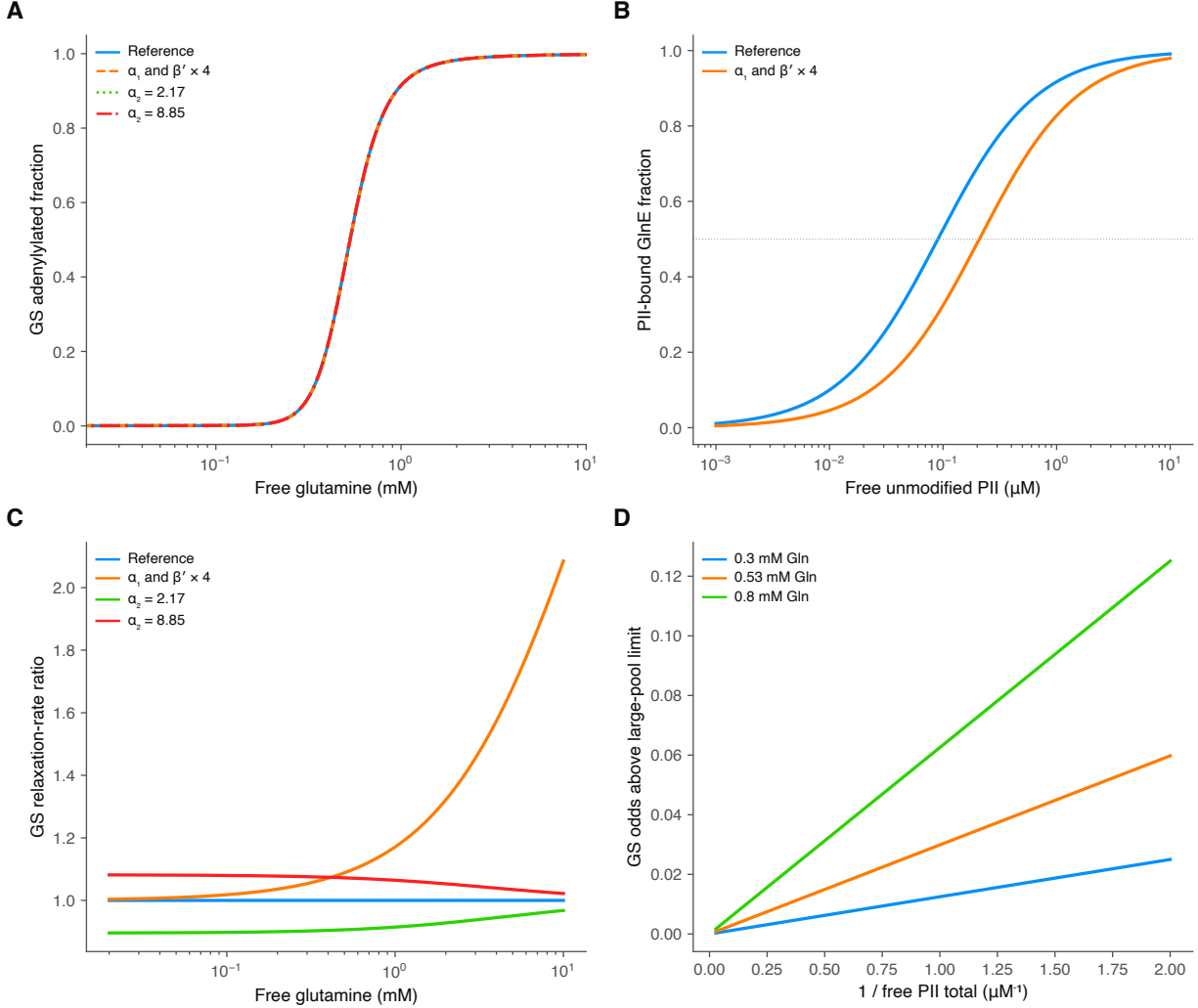


Figure S5: Binding and catalytic regulation in the published six-state GlnE topology. (A) Coupled GS steady-state curves are identical after a shared binding/catalysis transformation or a change in the PII–PII-UMP interaction parameter. One reference gain sets $Y = 0.5$ at 0.53 mM glutamine and $T = 5 \mu\text{M}$; it is unchanged in every variant. (B) PII-binding isotherms at fixed $G = 10$ mM and $U = 0.3 \mu\text{M}$ distinguish binding from catalytic synergy. (C) Reduced GS relaxation rates at separately controlled $P = 1 \mu\text{M}$ and $U = 0.3 \mu\text{M}$ differ along a glutamine titration. These are not the coupled inputs in A. (D) The PII-independent route gives a drive affine in inverse free-target total at fixed glutamine and modification fraction. Its analytic large-pool limit is subtracted to display the concentration-dependent term. Lines are predictions, not additional observations. A7 separates source-reported fitted constants from chosen parameters and states the rapid-binding and independent-GS assumptions.